Task 1

```
#include <iostream>
using namespace std;
int main( ){
    int num;
Cout << "Enter number";
  Cin >> num;
For (int i=1 ; i<=10; i++) {
  Cout << num << " * " << i << " =
}
    return 0;
}
```

output

Enter a ~~thi~~ive number: 5

Sum = 150

5x1 = 5
5x2 = 10
5x3 = 15
5x4 = 20
5x5 = 25
5x6 = 30
5x7 = 35
5x8 = 40
5x9 = 45
5x10 = 50

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter a number: ";
    cin >> num;

    for (int i = 1; i <= 10; i++)
    {
        cout << num << " x " << i << " = "
            << num * i << endl;
    }

    return 0;
}
```

```
Enter a number:
```

Task 2-

```
#include <iostream>
using namespace std;
int main(){
    int base, exponent;
    int power = 1;
  Cout << "Enter the base:";
    Cin >> base;
  Cout << " Enter the exponent";
    Cin >> exponent;
  If (exponent < 0){
  Cout << " please enter a non-negative exponent";
  } else
  {
    For (int i=1, i<= exponent; i++)
    {
      power = power *
base;
    }
  Cout << "power =" << pow;
  }  return 0;
}
```

output:-

```
Enter base=2
Enter exponent=4
power =16
```

task1.cxx task 2.cxx

```cpp
#include <iostream>
using namespace std;

int main()
{
    int base, exponent;
    int power = 1;

    cout << "Enter the base: ";
    cin >> base;

    cout << "Enter the exponent: ";
    cin >> exponent;

    if (exponent < 0)
    {
        cout << "Please enter a non-negative exponent.";
    }
    else
    {
        for (int i = 1; i <= exponent; i++)
        {
            power = power * base;
        }

        cout << "Power = " << power;
    }

    return 0;
}
```

TAB

Enter the base:

Task 3-

```
# include <iostream>
using namespace std;
    int main(){
    int n, sum=0;
Cout << "Enter n ";
 Cin >> n;
For (int i=1; i<=n; i++)  sum+=i;
cout << "sum = " << sum;
 return 0;
}
```

output:,

Enter a poritive number=5

Sum = 15

task3.cxx
/storage/emulate...

task 2.cxx task3.cxx

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n, sum = 0;

    cout << "Enter a positive number: ";
    cin >> n;

    if (n < 1)
    {
        cout << "Please enter a positive number.";
    }
    else
    {
        for (int i = 1; i <= n; i++)
        {
            sum = sum + i;
        }

        cout << "Sum = " << sum;
    }

    return 0;
}
```


TAB
Enter a positive number:

Task-4

```
#include <iostream>
using namespace std;
int main( ){
    int num, digit, sum=0;
Cout<<" Enter a number: ";
 Cin >> num;
 If (num <0)
 {
     num = -num;
  }
 while (num >0)
{
    digit = num % 10;
     Sum = Sum + digit;
     num= num /10;
   }
    Cout<<" Sum of digits= " <<Sum;
        return 0;
    }
```

output

```
Enter a number = 358
Sum of digits = 16
```

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num, digit, sum = 0;

    cout << "Enter a number: ";
    cin >> num;

    if (num < 0)
    {
        num = -num;
    }

    while (num > 0)
    {
        digit = num % 10;
        sum = sum + digit;
        num = num / 10;
    }

    cout << "Sum of digits = " << sum;

    return 0;
}
```


TAB
Enter a number:

Task 5-

```
#include <iostream>
using namespace;
int main() {
    int n;
    unsigned long factorial = 1;
Cout<<" Enter a number (0 to 20):";
    Cin>> n;
if (n<0 || n> 20) {
Cout <<please enter a number from
 0 to 20"; }
else
 {
    For (int i=1 ; i<=n ; i++)
{
    factorial = factorial * i;
}
Cout << " factorial = " << factorial;
 }
    return 0;
}
```

Output:-

```
Enter a number (0 to 20):4
Factorial = 24
```

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    unsigned long long factorial = 1;

    cout << "Enter a number (0 to 20): ";
    cin >> n;

    if (n < 0 || n > 20)
    {
        cout << "Please enter a number from 0 to 20.";
    }
    else
    {
        for (int i = 1; i <= n; i++)
        {
            factorial = factorial * i;
        }

        cout << "Factorial = " << factorial;
    }

    return 0;
}
```


TAB
Enter a number (0 to 20):

Task - 6

```
#include <iostream>
using namespace std;
int main( ) {
    for (int i=1; i<= 6; i++)
  {
    for (int j= 1; j <= i ; j++ )
    {
      cout << " *" ;
    }
      cout<< endl;
    }
    return 0;
  }
```

output:

```
 *
 * *
 * * *
 * * * *
 * * * * *
 * * * * * *
```

```cpp
#include <iostream>
using namespace std;

int main()
{
    for (int i = 1; i <= 6; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            cout << "*";
        }

        cout << endl;
    }

    return 0;
}
```

TAB

```
*
**
***
****
*****
******

[Program finished]
```

```
    case 10:
cout<<" October: 31 days"; break;
    case 11:
cout<<" November: 30 days"; break;
    case 12:
cout<<" December: 31 days"; break;
    default:
cout<<"Invalid month number!";
    return 0;
}
```

output:

```
Enter month number (1-12): 4
April: 30 days
```

Task 7-

```
#include <iostream>
using namespace std;
int main(){
    int month;
    Cout << Enther month number (1-12): ";
    Cin >> month;
    swith (month){
        case1:
    Cout << " January : 31 days ";
        break;
        case2:
    Cout << " February : 28 oo 29 days "; break;
        case 3:
    Cout<< " March : 31 days "; break;
        case 4:
    Cout << " April : 30 days "; break;
        case 5:
    Cout << " May : 31 days "; ; break;
        case6:
    Cout << " June: 30 days "; break;
        case 7:
    Cout<< " July : days 31 "; break;
        case 8:
    Cout<< " August : days 31 "; break;
        case 9:
    Cout<< " reptember: 30 days ";
```

```cpp
#include <iostream>
using namespace std;

int main()
{
    int month;

    cout << "Enter month number (1-12): ";
    cin >> month;

    switch (month)
    {
        case 1:
            cout << "January: 31 days";
            break;

        case 2:
            cout << "February: 28 or 29 days";
            break;

        case 3:
            cout << "March: 31 days";
            break;

        case 4:
            cout << "April: 30 days";
            break;

        case 5:
            cout << "May: 31 days";
            break;
```

task 2.cxx task3.cxx task4.cxx task5.cxx ta

```
33          case 6:
34              cout << "June: 30 days";
35              break;
36
37          case 7:
38              cout << "July: 31 days";
39              break;
40
41          case 8:
42              cout << "August: 31 days";
43              break;
44
45          case 9:
46              cout << "September: 30 days";
47              break;
48
49          case 10:
50              cout << "October: 31 days";
51              break;
52
53          case 11:
54              cout << "November: 30 days";
55              break;
56
57          case 12:
58              cout << "December: 31 days";
59              break;
60
61          default:
62              cout << "Invalid month number!";
63      }
64
65      return 0;
66  }
```




TAB
Enter month number (1-12):

Task - 8

```
#include <iostream>
using namespace std;
int main() {
    char ch;
    cout << " Enter an alphabet : ";
    cin >> ch;
    switch (ch)
    {
        case 'a' :
        case 'e' :
        case 'i' :
        case 'o' :
        case 'u' :
        case 'A' :
        case 'E' :
        case 'I' :
        case 'O' :
        case 'U' ;
    cout << " It is a Vowel. ";
            break ;
        default:
    if ((ch >= 'a' && ch <= 'z') ||
    (ch >= 'A' && ch <= 'z') )
    }
        else
        {
    cout << " Invalid input ";
    }
        return 0;
}
```

output:1

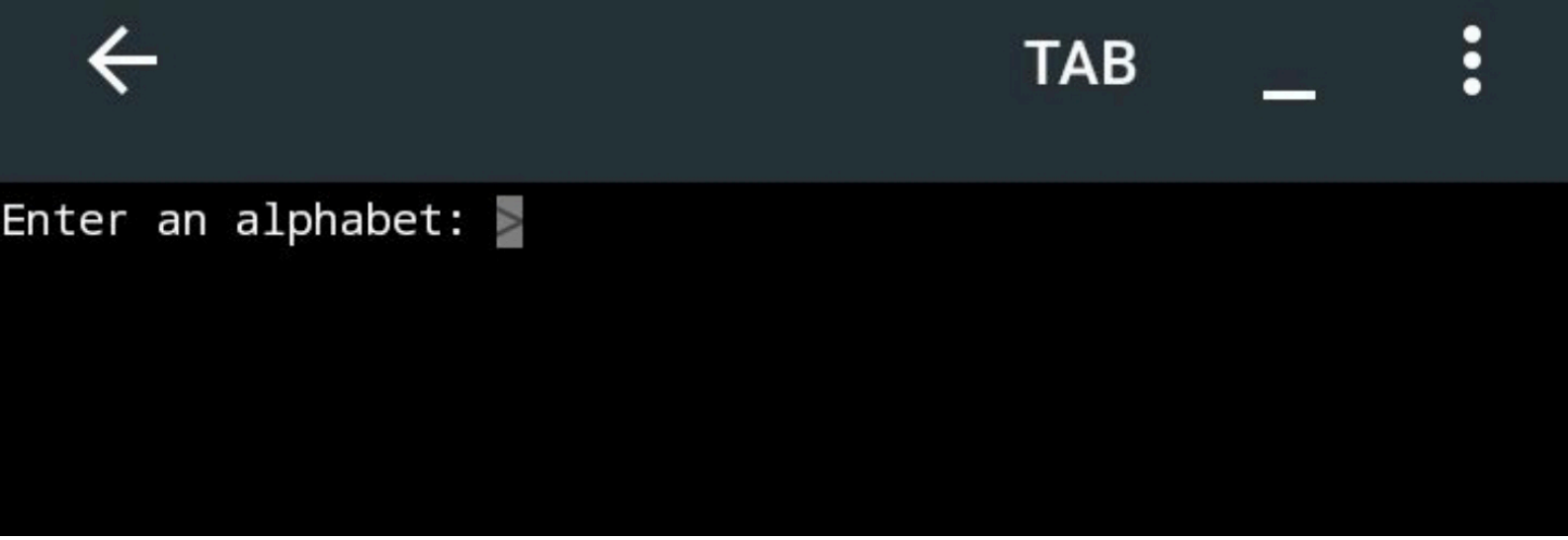
Enter an alphabet : A
It is a vowel

out put 2.

Enter an alphabet : b
It is a consonant.

```cpp
#include <iostream>
using namespace std;
int main()
{
    char ch;
    cout << "Enter an alphabet: ";
    cin >> ch;
    switch (ch)
    {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
        case 'A':
        case 'E':
        case 'I':
        case 'O':
        case 'U':
            cout << "It is a vowel.";
            break;

        default:
            if ((ch >= 'a' && ch <= 'z') ||
                (ch >= 'A' && ch <= 'Z'))
            {
                cout << "It is a consonant.";
            }
            else
            {
                cout << "Invalid input!";
            }
    }
```


TAB
Enter an alphabet: